

AUSTIN, MN, USA

WMO: 727566

Lat: 43.665N

Lon: 92.933W

Elev: 1233

StdP: 14.05

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04902

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.4	-7.8	-18.3	2.1	-10.1	-15.0	2.6	-7.2	27.8	16.5	25.6	16.1	7.5	300	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	89.7	74.5	86.0	72.7	82.5	70.6	77.9	84.9	75.5	82.7	73.7	80.7	11.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
0.4%			1%			2%			0.4%		1%		2%			
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		(p)
75.3	138.8	82.1	73.0	128.2	80.2	71.9	123.3	79.1	42.2	85.3	40.0	82.1	38.1	80.6		86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.4	20.8	18.7	-18.8	93.5	6.0	3.3	-23.1	95.9	-26.7	97.8	-30.0	99.7	-34.4	102.0		
			-19.0	80.5	5.7	2.9	-23.1	82.7	-26.4	84.4	-29.7	86.0	-33.8	88.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	15.7	19.5	32.4	46.0	58.0	68.2	72.1	69.2	62.2	49.2	34.8	21.8
	DBStd	21.99	13.27	12.63	12.05	9.99	8.55	6.51	5.34	5.37	8.32	9.25	11.01	11.81
	HDD50	4170	1065	854	561	188	23	0	0	0	8	129	467	875
	HDD65	7596	1530	1274	1012	573	248	41	7	21	150	495	905	1340
	CDD50	2669	0	0	14	68	273	545	684	594	374	105	12	0
	CDD65	621	0	0	0	3	33	136	227	149	67	6	0	0
	CDH74	5219	0	0	2	48	337	1134	1945	1116	567	69	1	0
	CDH80	1473	0	0	0	11	100	328	606	271	142	15	0	0
Wind	WSAvg	8.3	9.3	9.3	9.4	10.1	9.1	7.4	6.2	5.6	6.8	8.5	8.9	8.8
Precipitation	PrecAvg	35.9	0.9	0.9	2.0	3.7	4.4	5.4	4.9	4.9	3.4	2.4	1.9	1.3
	PrecMax	43.3	2.0	2.0	3.5	7.3	8.0	9.3	11.2	11.7	9.6	7.7	5.9	3.1
	PrecMin	25.4	0.2	0.2	0.3	1.1	1.8	2.2	1.1	1.3	1.0	0.6	0.1	0.2
	PrecStd	5.0	0.5	0.5	1.0	1.8	1.7	1.9	2.2	2.3	2.2	1.6	1.5	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	52.0	70.5	81.0	89.6	91.3	93.1	90.2	89.7	81.7	68.0	53.9
		MCWB	38.9	46.5	58.4	61.9	69.5	74.4	77.5	75.2	72.3	64.8	56.0	49.3
	2%	DB	38.9	44.7	62.0	72.6	81.8	87.8	89.9	85.8	83.8	73.3	61.3	45.1
		MCWB	35.1	40.9	54.7	57.0	64.6	72.1	76.7	74.5	70.7	60.8	52.5	41.1
	5%	DB	36.1	39.1	54.8	66.5	77.1	83.6	86.2	82.3	81.0	69.7	55.0	39.3
		MCWB	33.5	36.0	49.0	54.0	62.9	69.6	74.2	72.3	68.2	59.8	48.5	36.3
	10%	DB	33.8	36.3	48.1	62.9	72.7	81.0	82.4	80.9	76.8	63.8	50.4	36.5
		MCWB	32.0	34.0	44.0	51.7	60.4	68.1	71.9	70.8	66.3	55.8	44.9	34.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	47.2	61.3	63.4	73.0	78.0	82.3	80.3	76.3	68.7	58.6	49.8
		MCDB	44.5	49.9	68.1	74.2	83.2	86.5	88.5	85.3	83.4	76.4	65.4	53.9
	2%	WB	35.9	40.8	55.6	59.4	68.4	74.9	78.5	76.8	73.1	64.3	53.8	41.3
		MCDB	38.7	44.3	60.9	69.1	77.9	83.3	85.5	83.0	80.7	69.2	59.5	44.6
	5%	WB	33.5	36.8	49.7	56.1	65.7	72.7	76.3	74.5	70.5	60.5	49.3	37.3
		MCDB	35.3	38.9	55.2	65.2	74.0	80.9	83.7	80.6	77.7	67.3	55.3	39.5
	10%	WB	31.3	33.8	44.1	52.8	62.7	70.4	74.3	72.4	67.8	56.7	45.5	34.6
		MCDB	33.2	36.1	48.4	60.8	69.8	78.4	81.3	78.2	75.1	63.4	49.4	36.3
Mean Daily Temperature Range	5% DB	MDBR	14.9	15.4	15.4	18.7	19.2	18.4	18.7	18.5	20.5	18.6	15.3	13.5
		MCDBR	16.9	16.3	23.3	27.8	25.0	22.4	21.2	20.5	23.8	26.0	22.8	16.9
		MCWBR	14.4	13.8	16.2	15.8	12.5	10.8	11.0	11.5	12.6	14.6	15.5	14.3
	5% WB	MCDBR	15.6	15.0	22.6	24.2	20.5	19.2	18.9	17.9	20.4	20.8	21.1	15.2
		MCWBR	13.6	13.0	16.4	15.5	12.5	11.4	11.8	11.8	12.4	14.5	15.1	13.2
Clear-Sky Solar Irradiance	taub	0.266	0.291	0.314	0.365	0.392	0.410	0.415	0.403	0.371	0.326	0.289	0.265	
	taud	2.418	2.341	2.410	2.315	2.300	2.289	2.269	2.305	2.366	2.483	2.506	2.448	
	Ebn at Noon	278	287	294	285	279	273	271	270	271	272	268	268	
	Edh at Noon	23	30	32	39	41	41	42	39	34	26	22	21	
All-Sky Solar Radiation	RadAvg	563	897	1188	1461	1720	1919	2002	1692	1353	878	578	424	
	RadStd	61	94	120	151	131	146	89	117	96	118	50	34	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-1.35	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page